Data Storytelling

Creating Better Visualizations with examples in R and Python

Kenneth C. Enevoldsen

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Agenda

How to make better visualisations

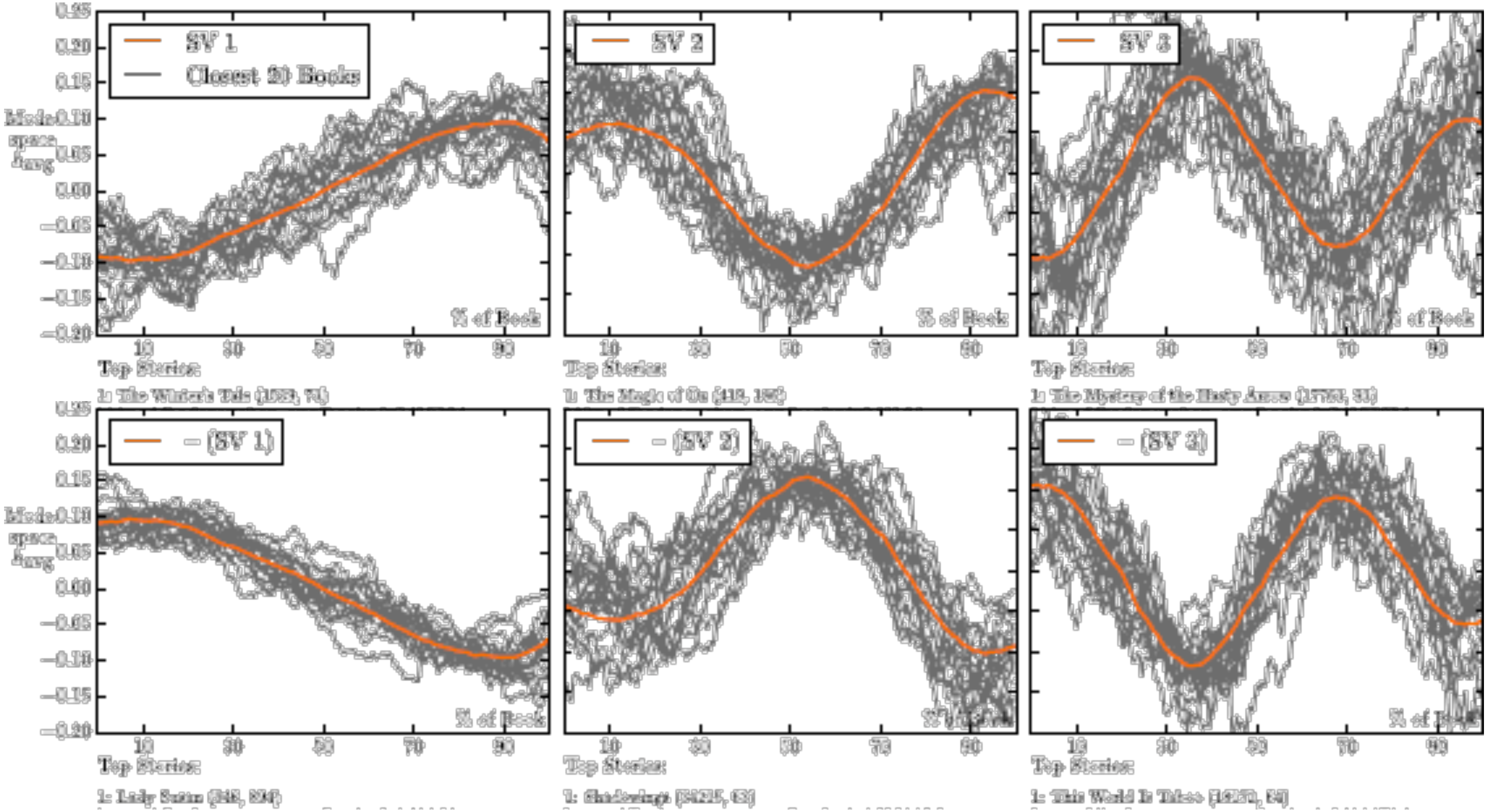
- Data storytelling
- Open-Science Visualizations
- Interactivity

Data Storytelling

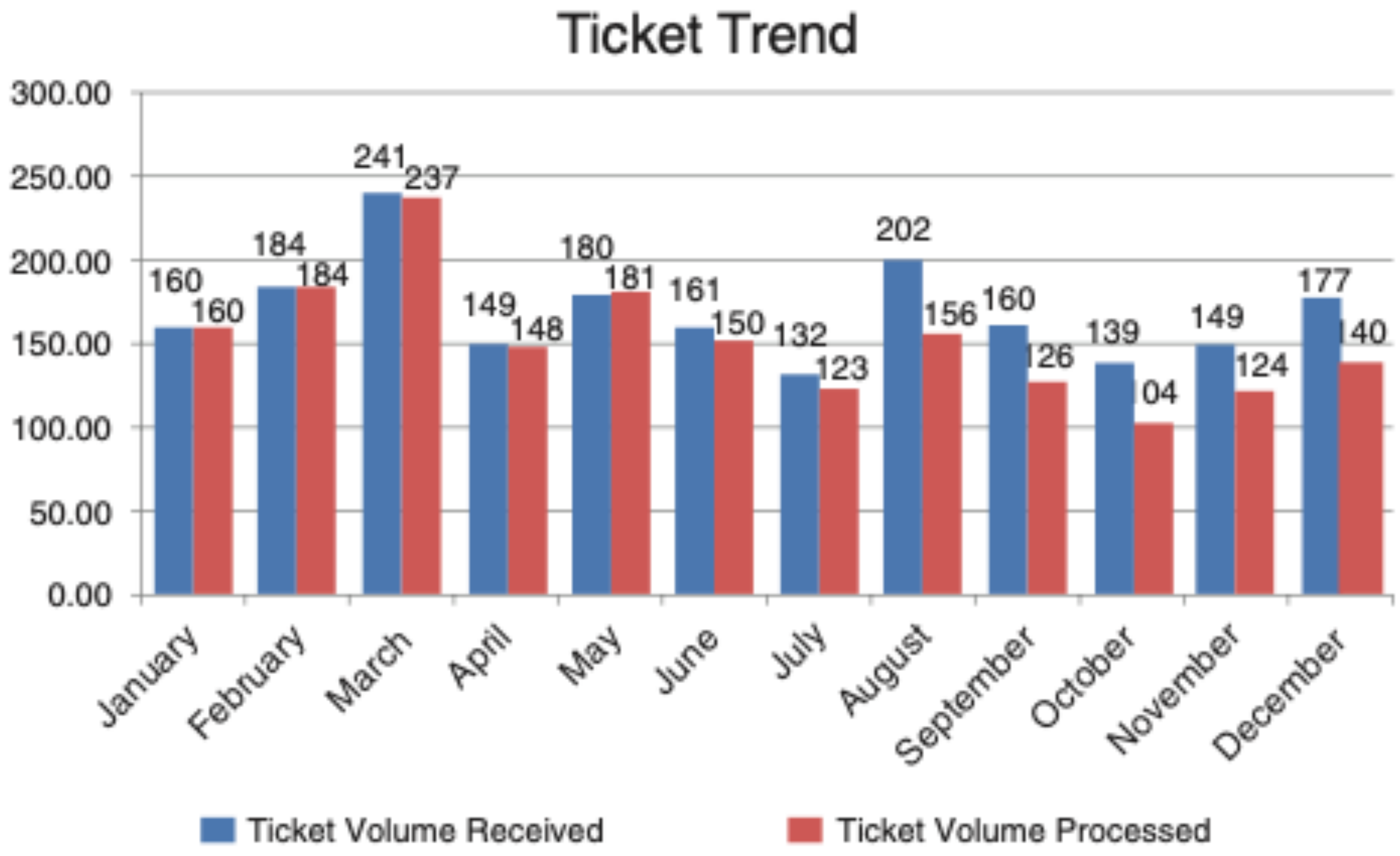
Good visualisations tells stories

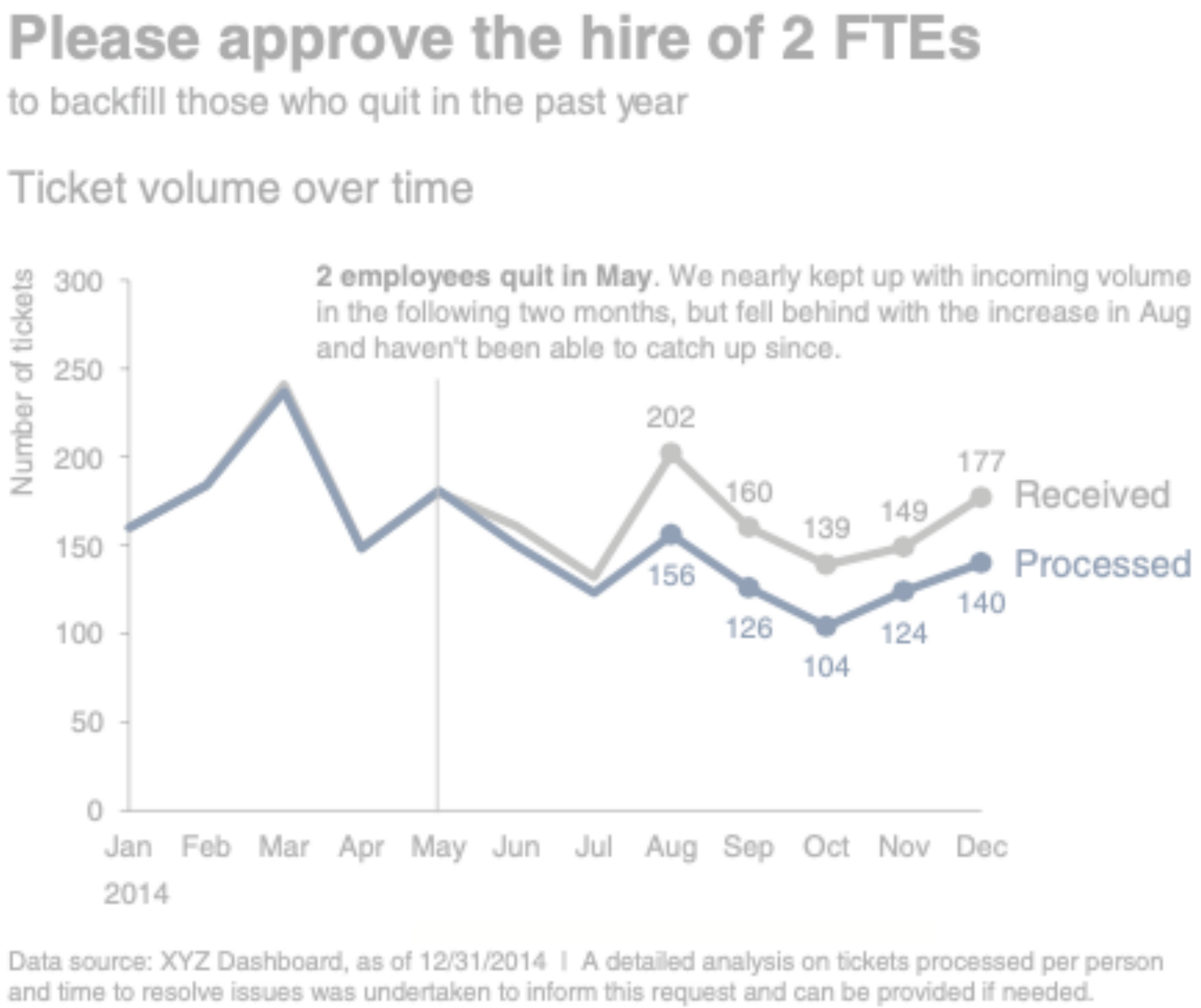
- 90% remember the highlight of red riding hood

Source: Storytelling with Data



Source: The emotional arcs of stories are dominated by six basic shapes



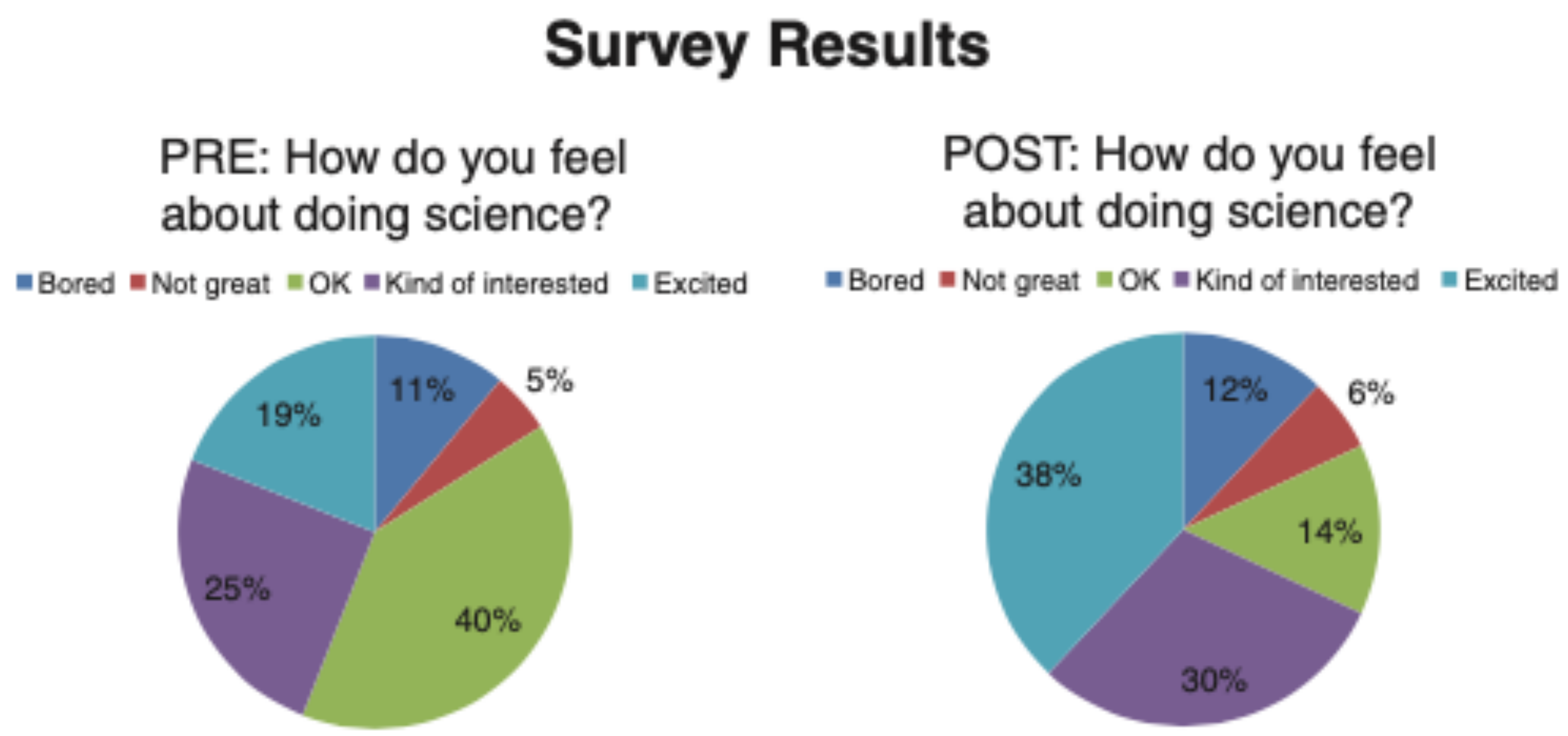


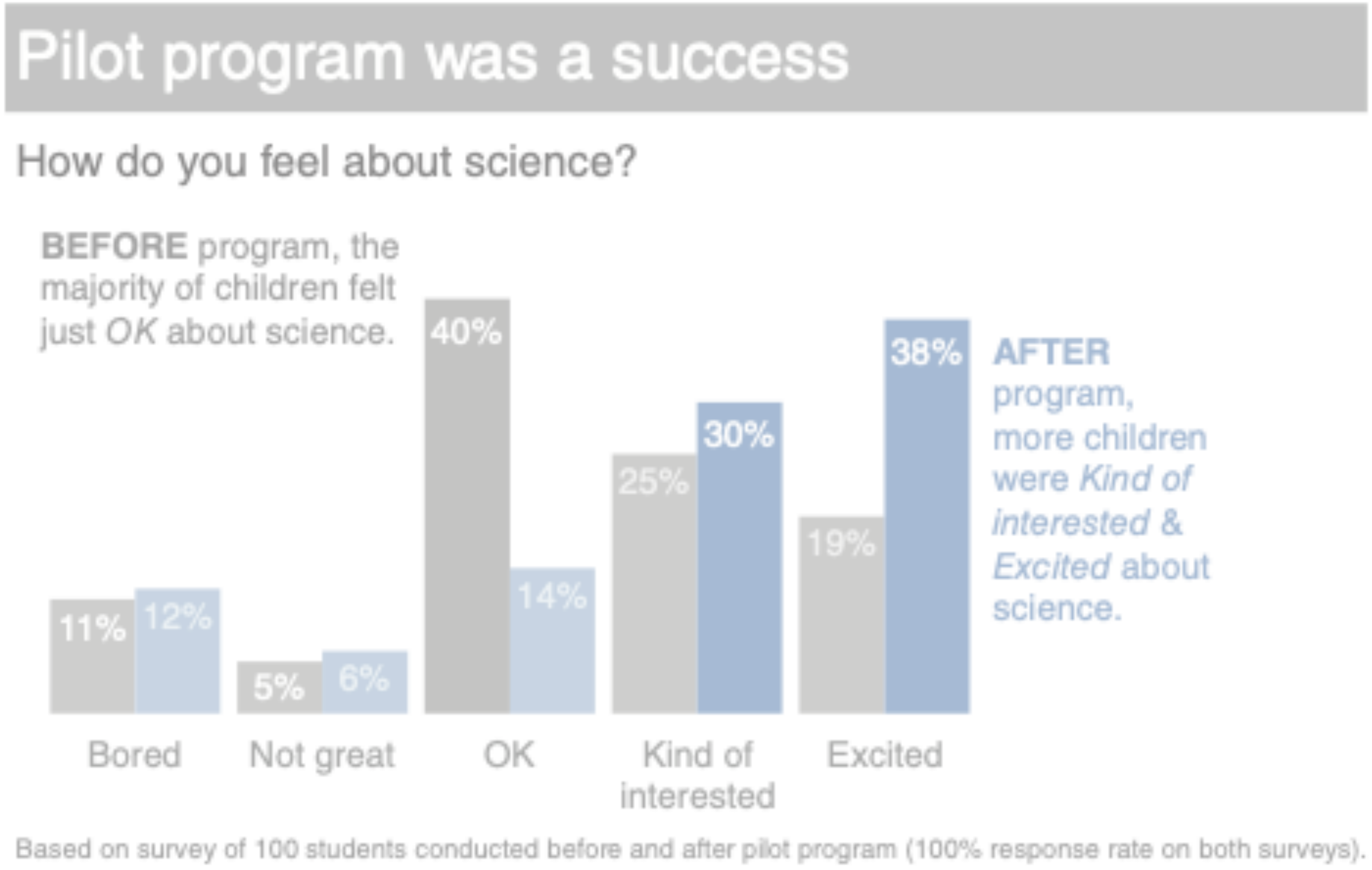
Point 1: Only plot exact values when relevant

Point 2: Change over time is better visualised with lines

Point 3: Use plot annotations to convey story

Source: Storytelling with Data



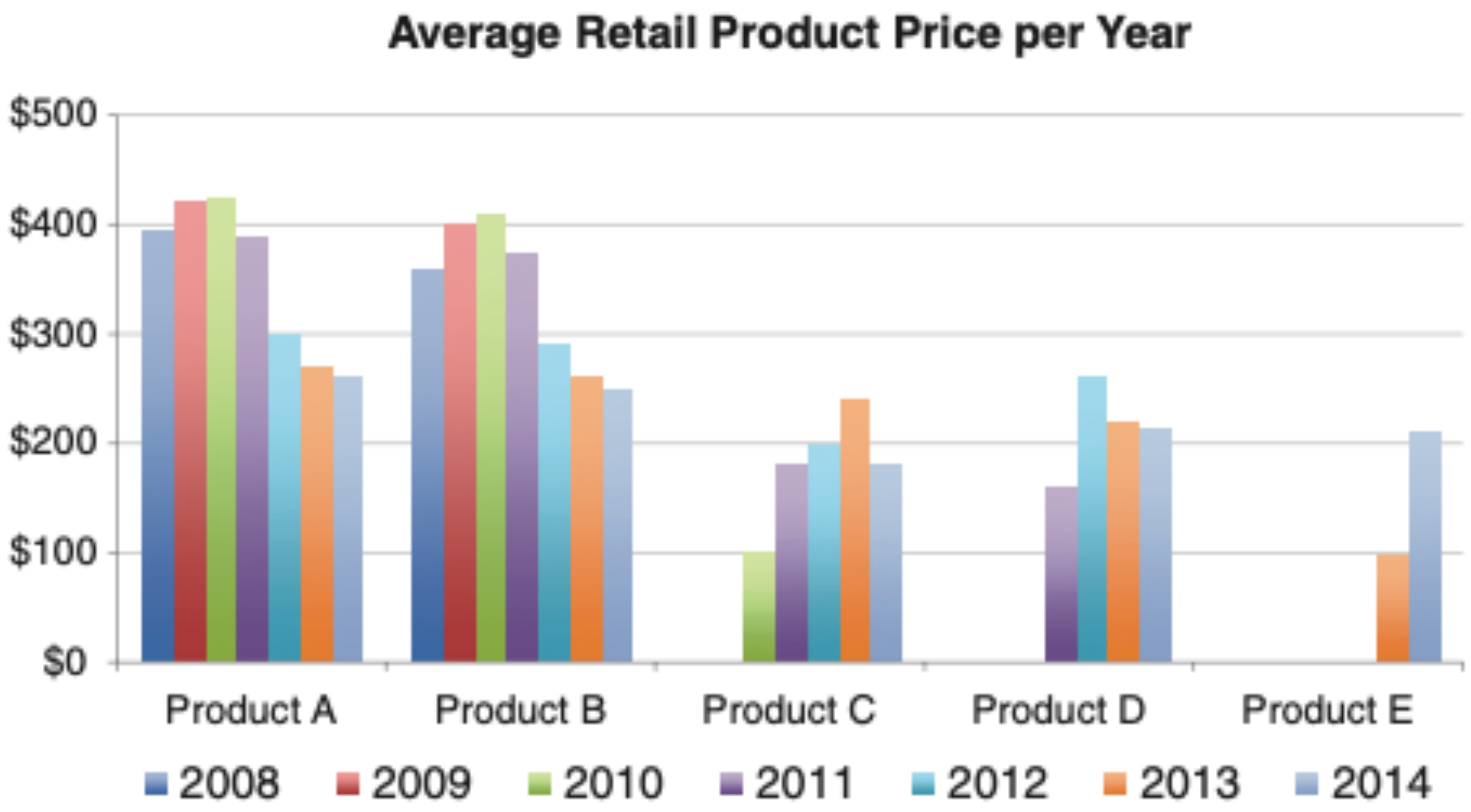


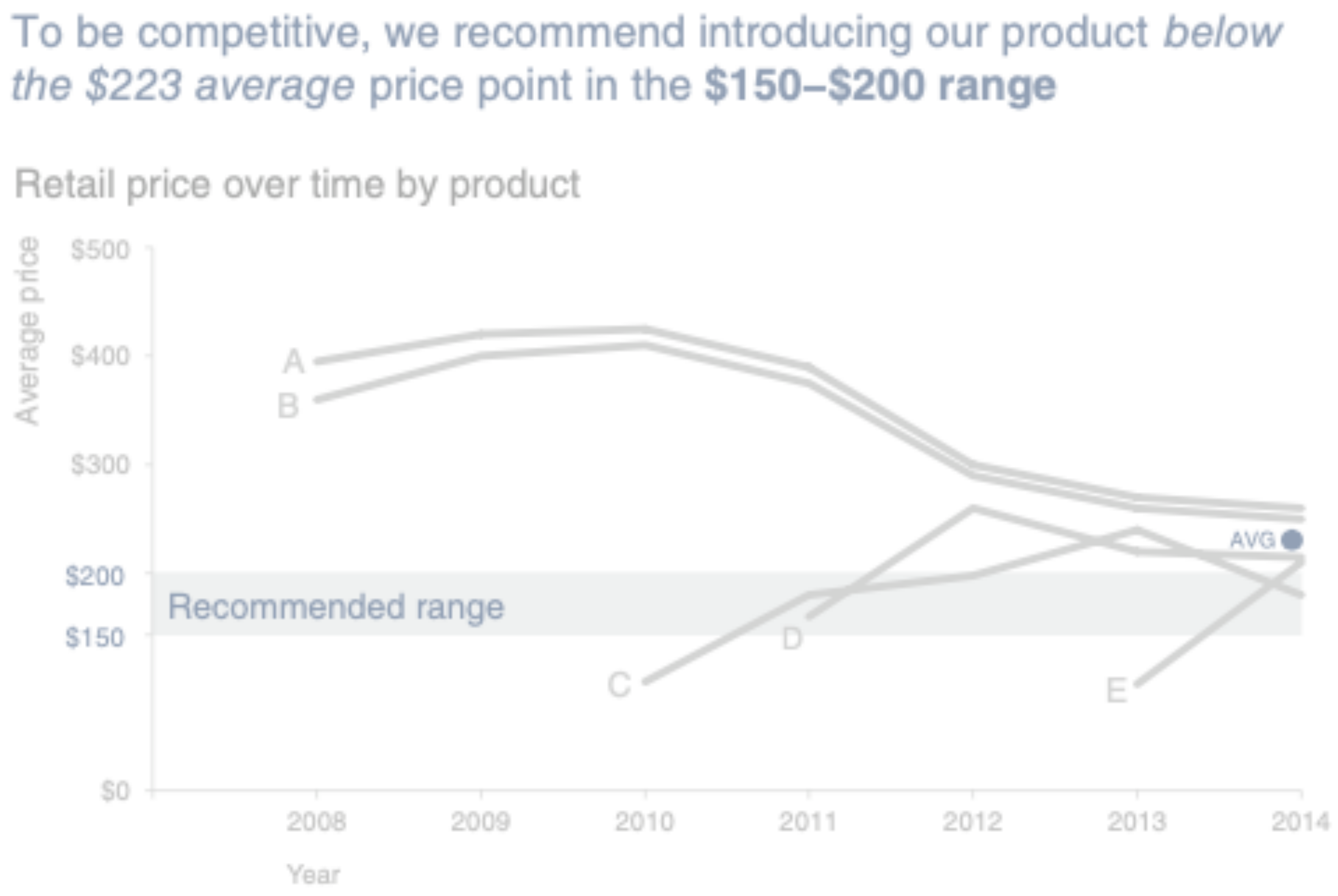
Point 4: Make datapoints which should be compared, comparable

Point 5: Use shading to guide attention

Point 6: Make conclusion from the plot clear

Source: Storytelling with Data





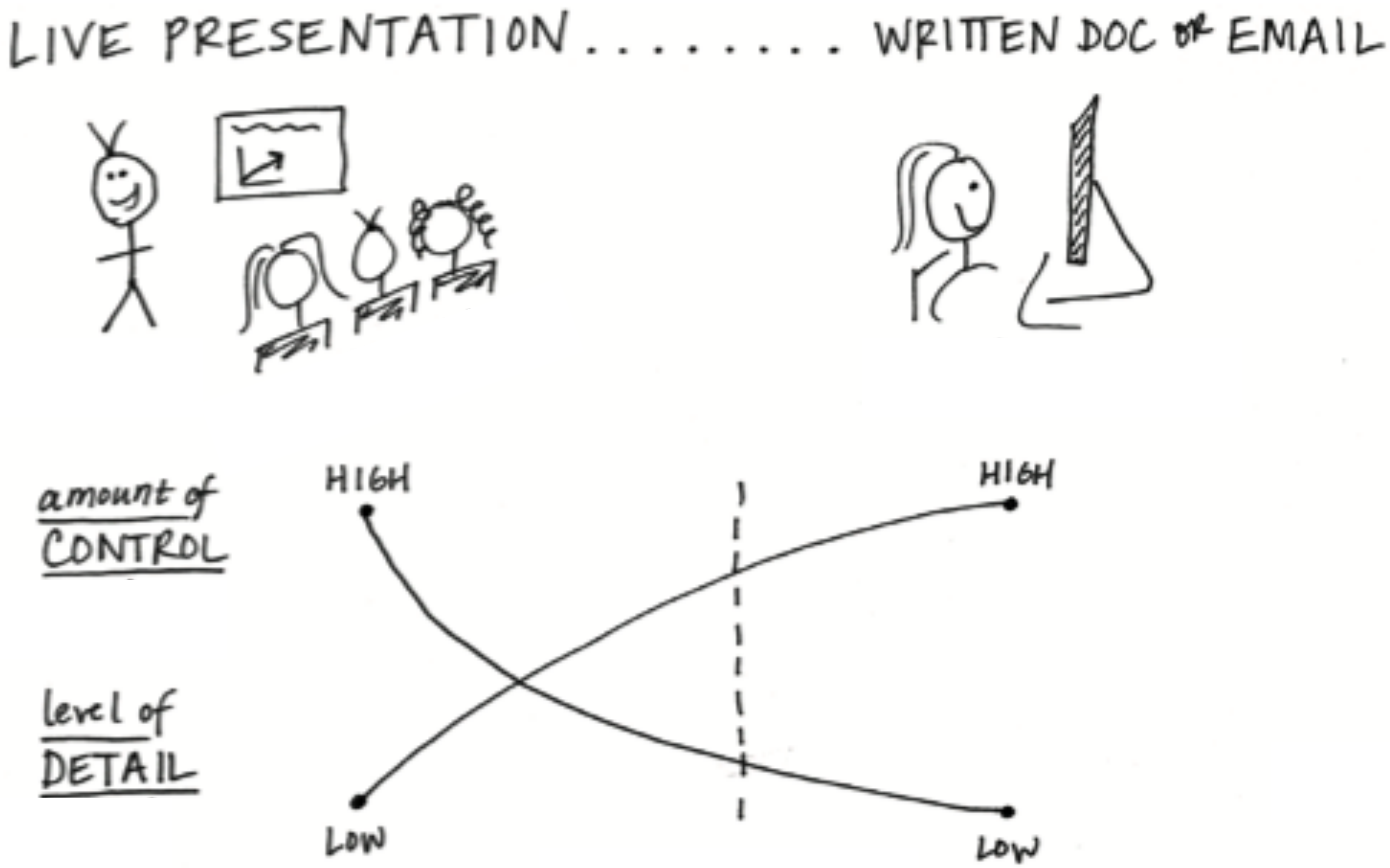
Point 4: Make datapoints which should be compared, comparable

Point 2: Change over time is better visualised with lines. **Categories should generally be labels, no an axis**

Point 5: Use shading to guide attention, **e.g. by highlighting the x-axis**

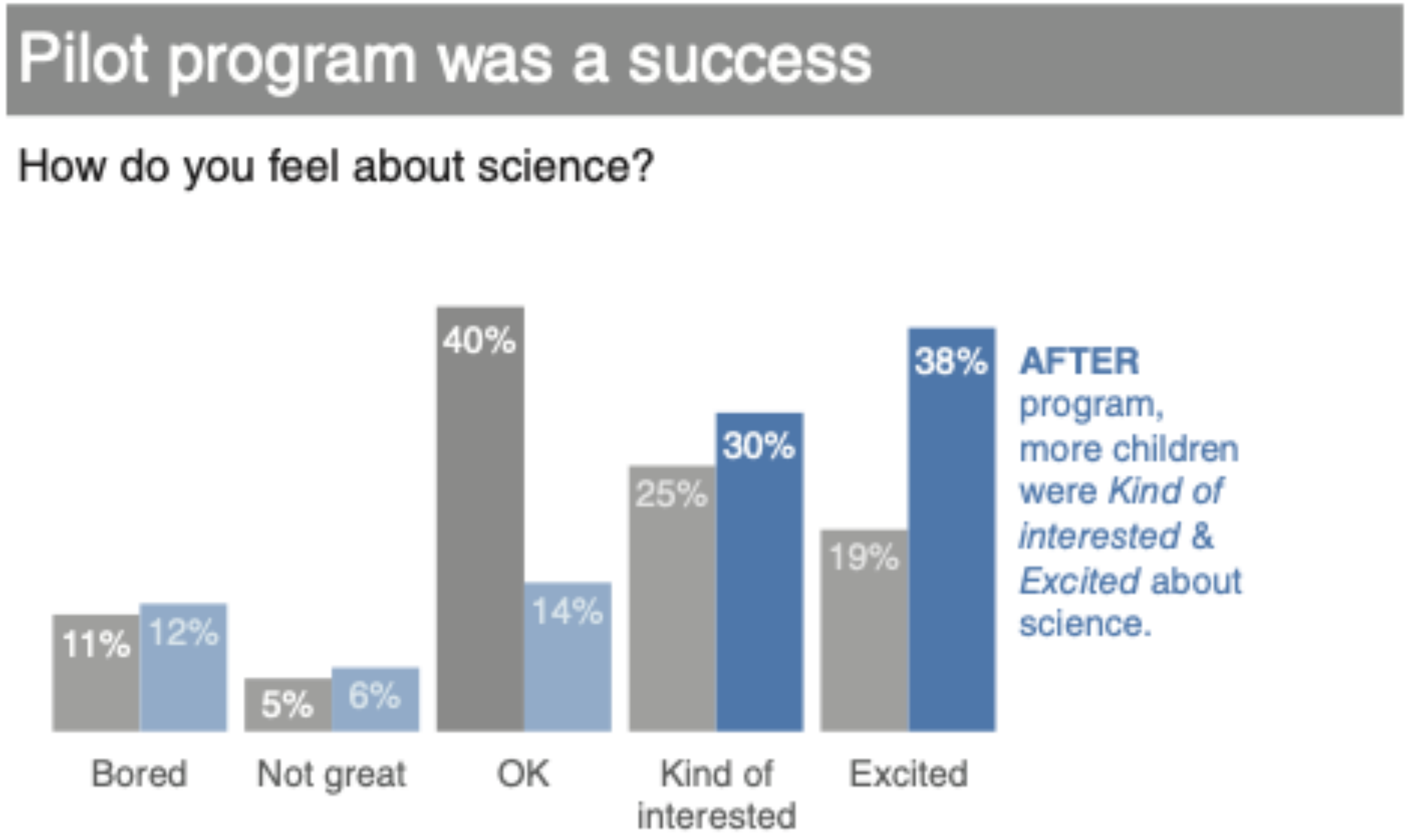
Point 6: Make conclusion from the plot clear, **e.g. using annotations**

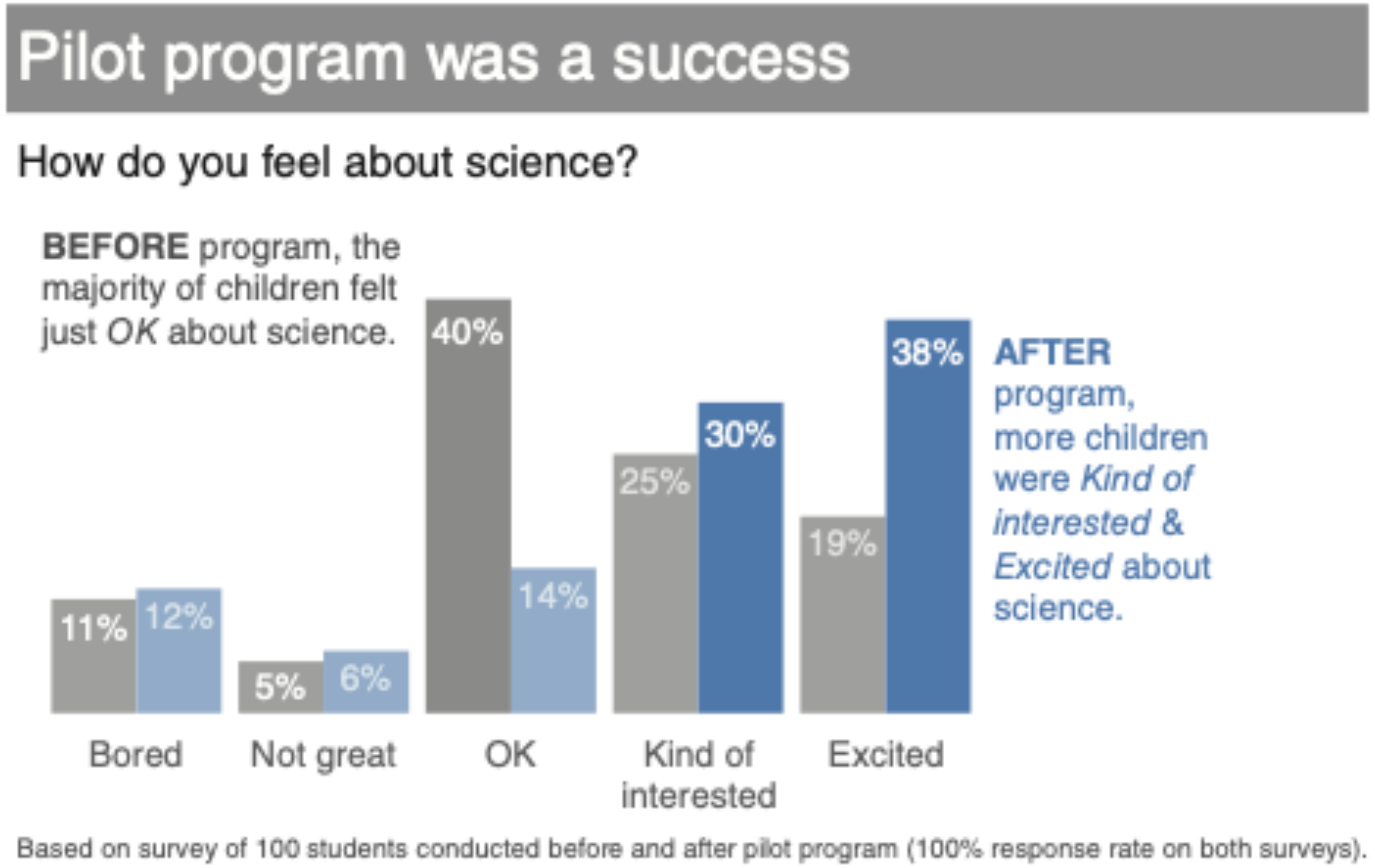
Source: Storytelling with Data



Source: Storytelling with Data

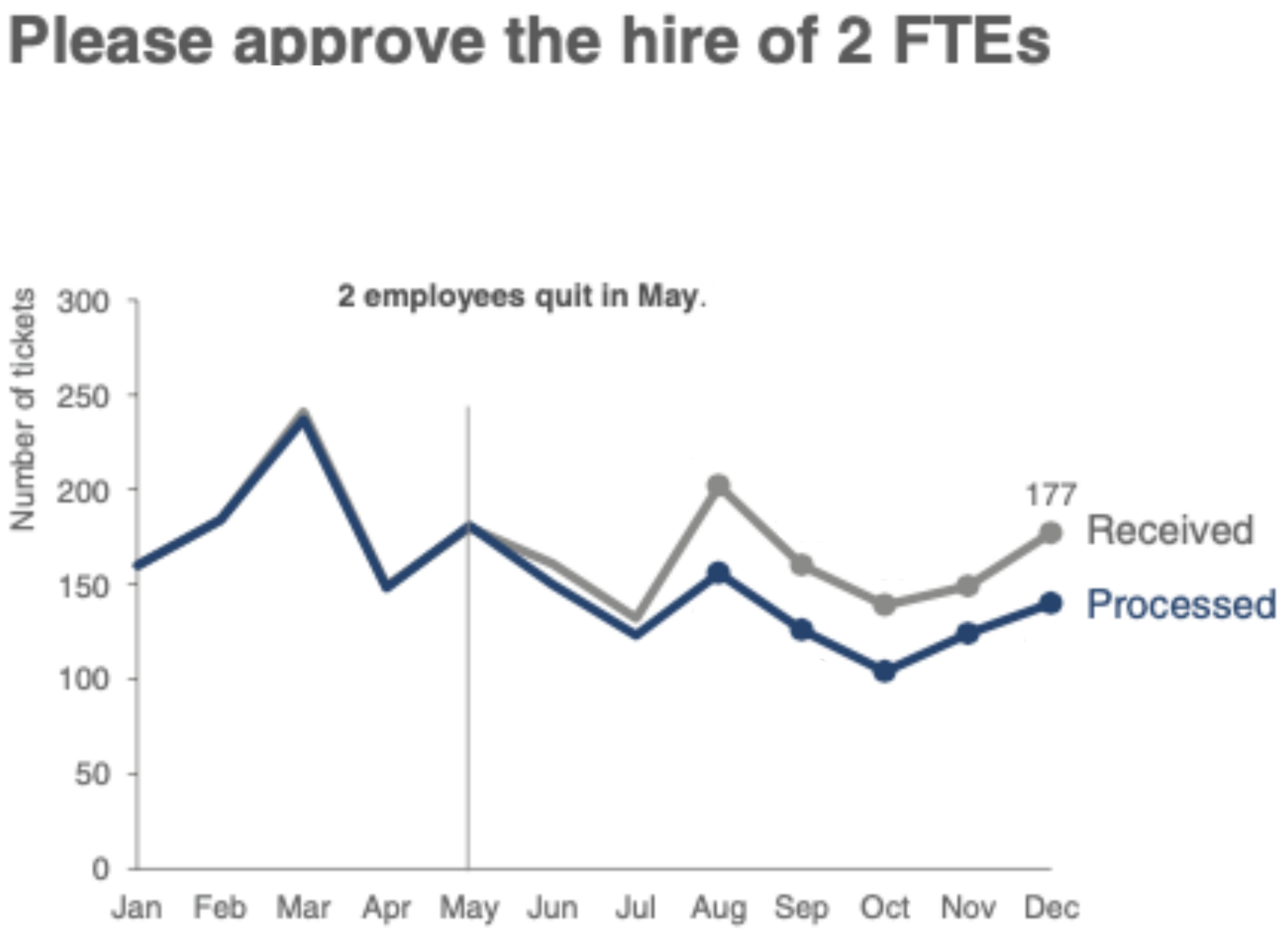
LIVE PRESENTATION . . . . WRITTEN DOC or EMAIL

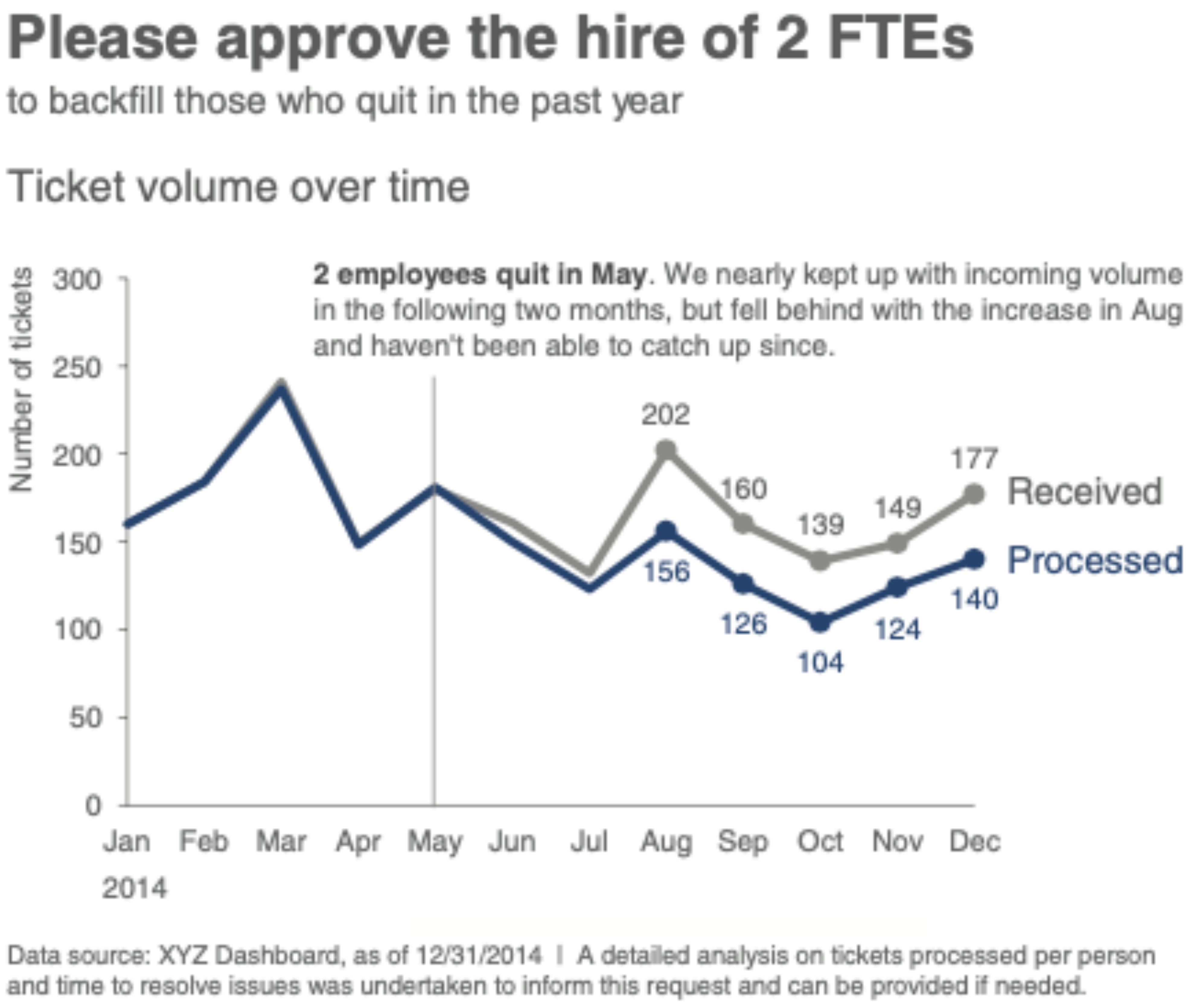




Source: Storytelling with Data

LIVE PRESENTATION . . . . WRITTEN DOC OR EMAIL





Source: Storytelling with Data

Data Storytelling

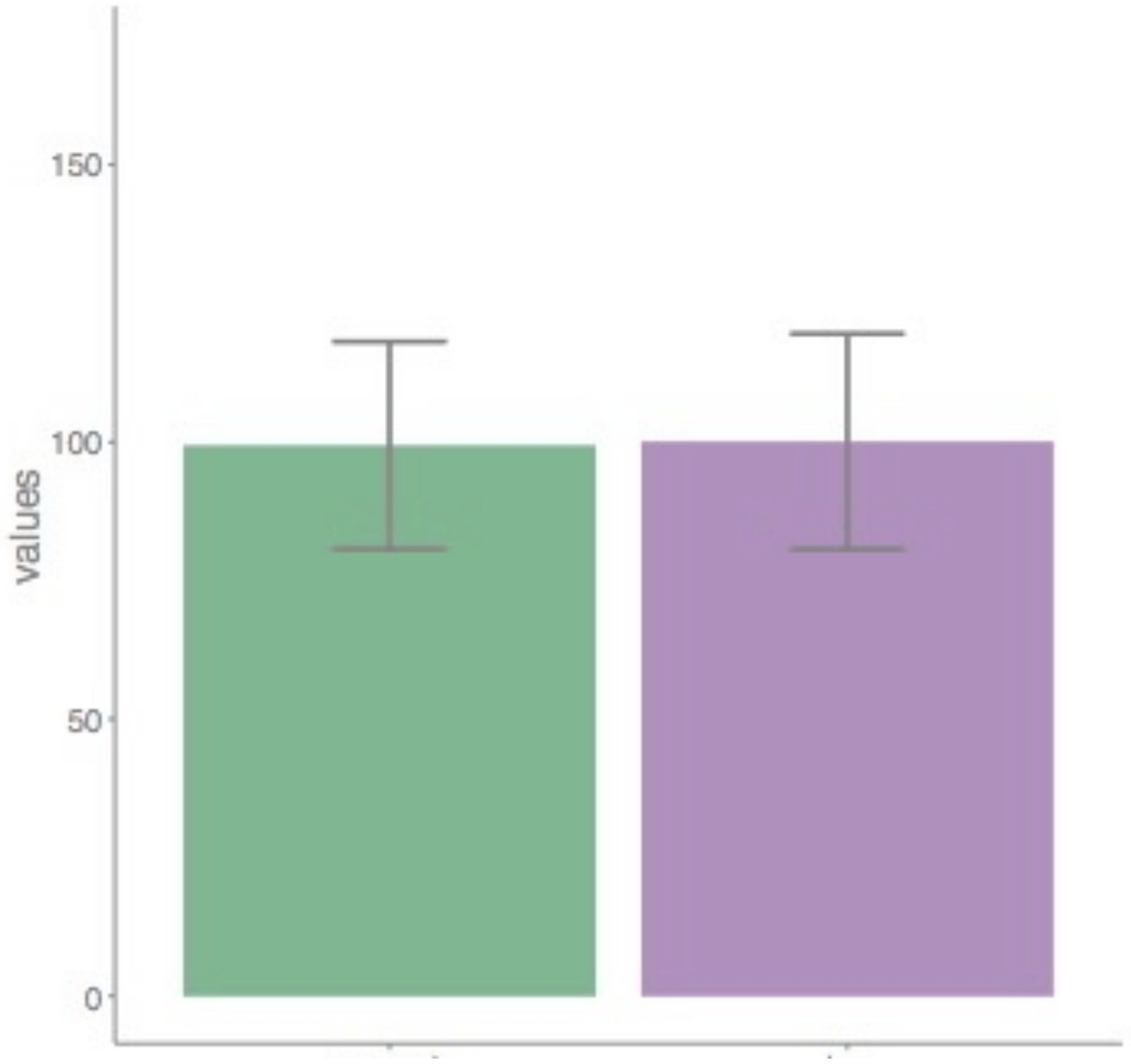
Good visualisations tells stories

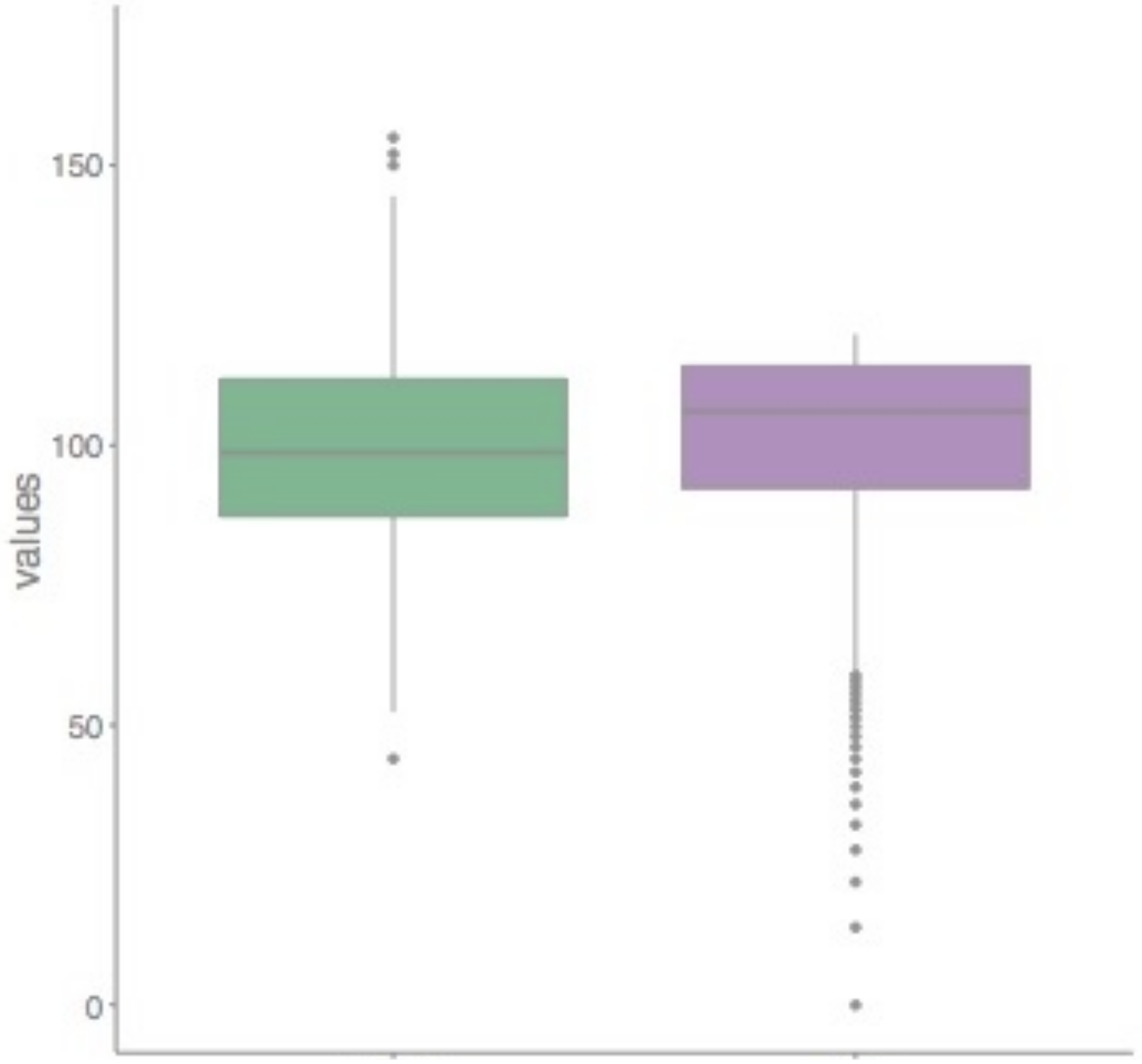
1. Goal directed, say what you meant to say
2. Keep it simple
3. Be sympathetic to your reader

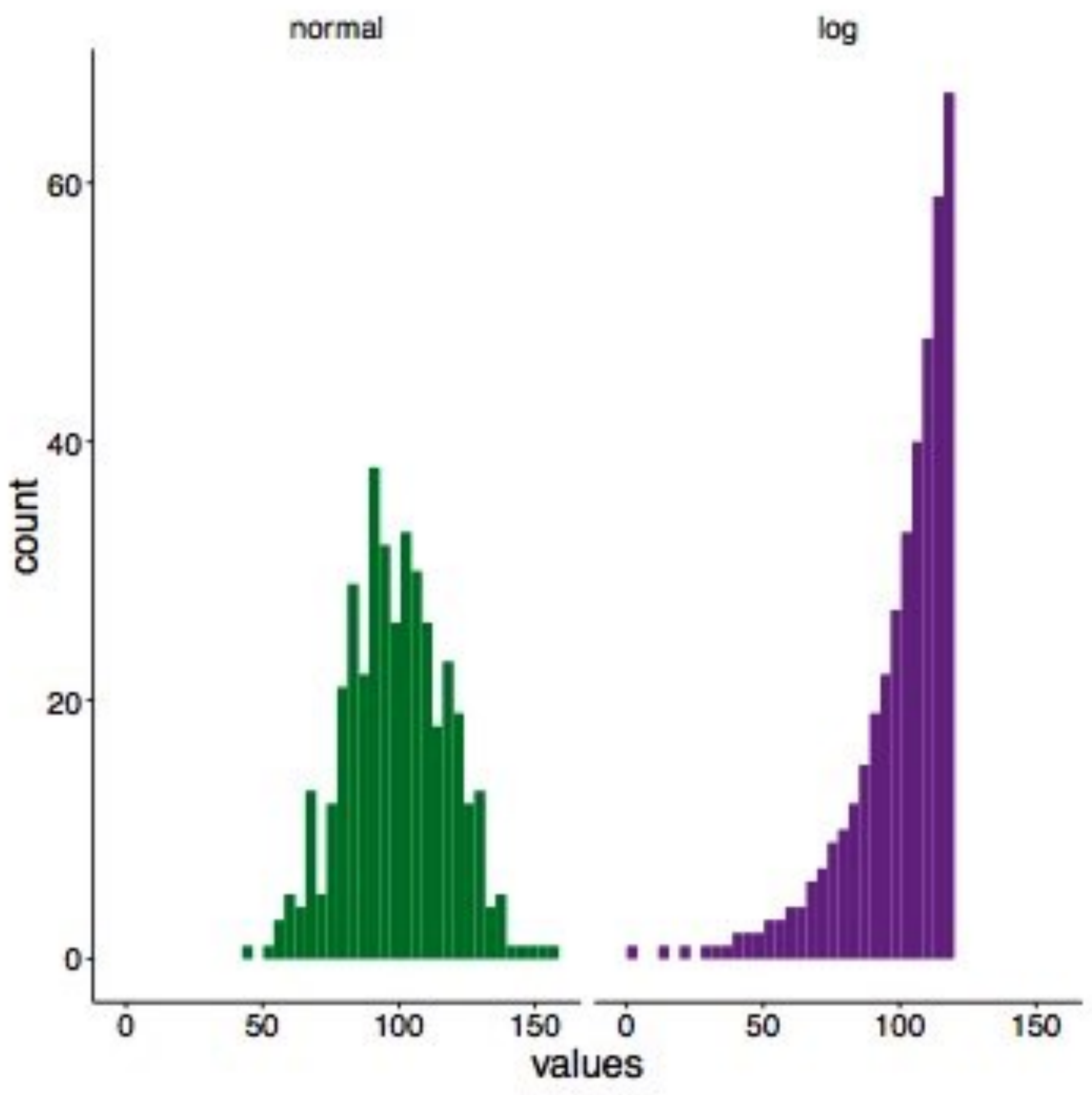
Open-Science in Visualisations

When visualisations is an ethics statement

- Don’t hide relevant information
- Make convincing visualizations
- Should allow the user to question the data



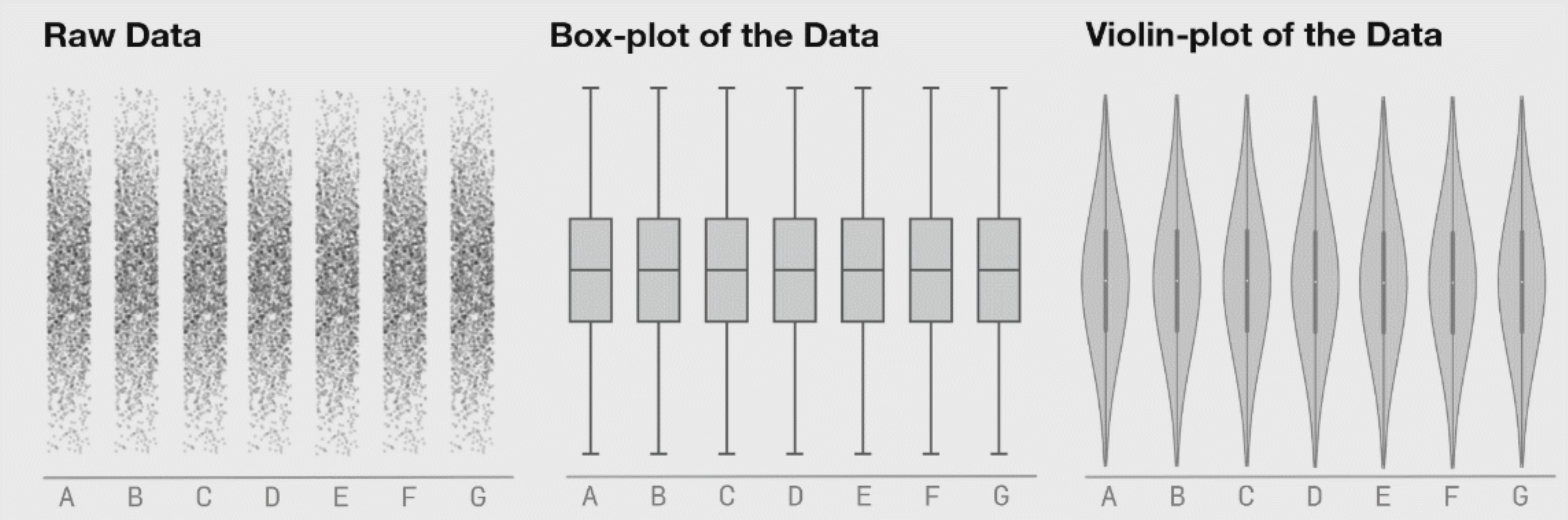




Point 8: Bar plots can hide distribution of data

Point 5: Use shading to guide attention

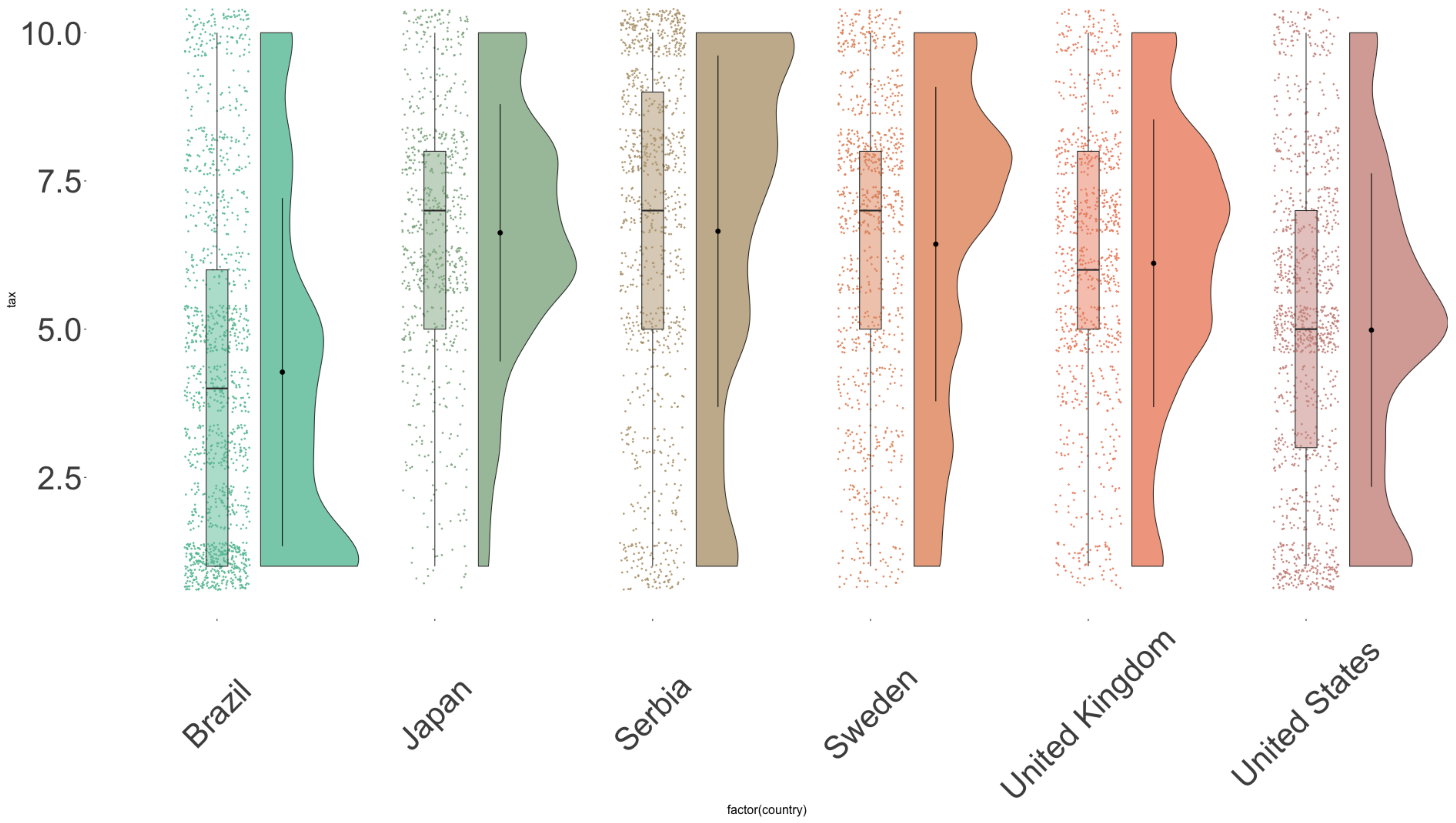
Source: #BarBarPlots!



What is the problem with Violins?

Point 8.1: **Box plots** can hide data distribution

Point 9: bar plots, box plots and violin plots hide meaningful information in regards to count



What is problem does this plot have?

Source: Introducing Raincloud Plots!

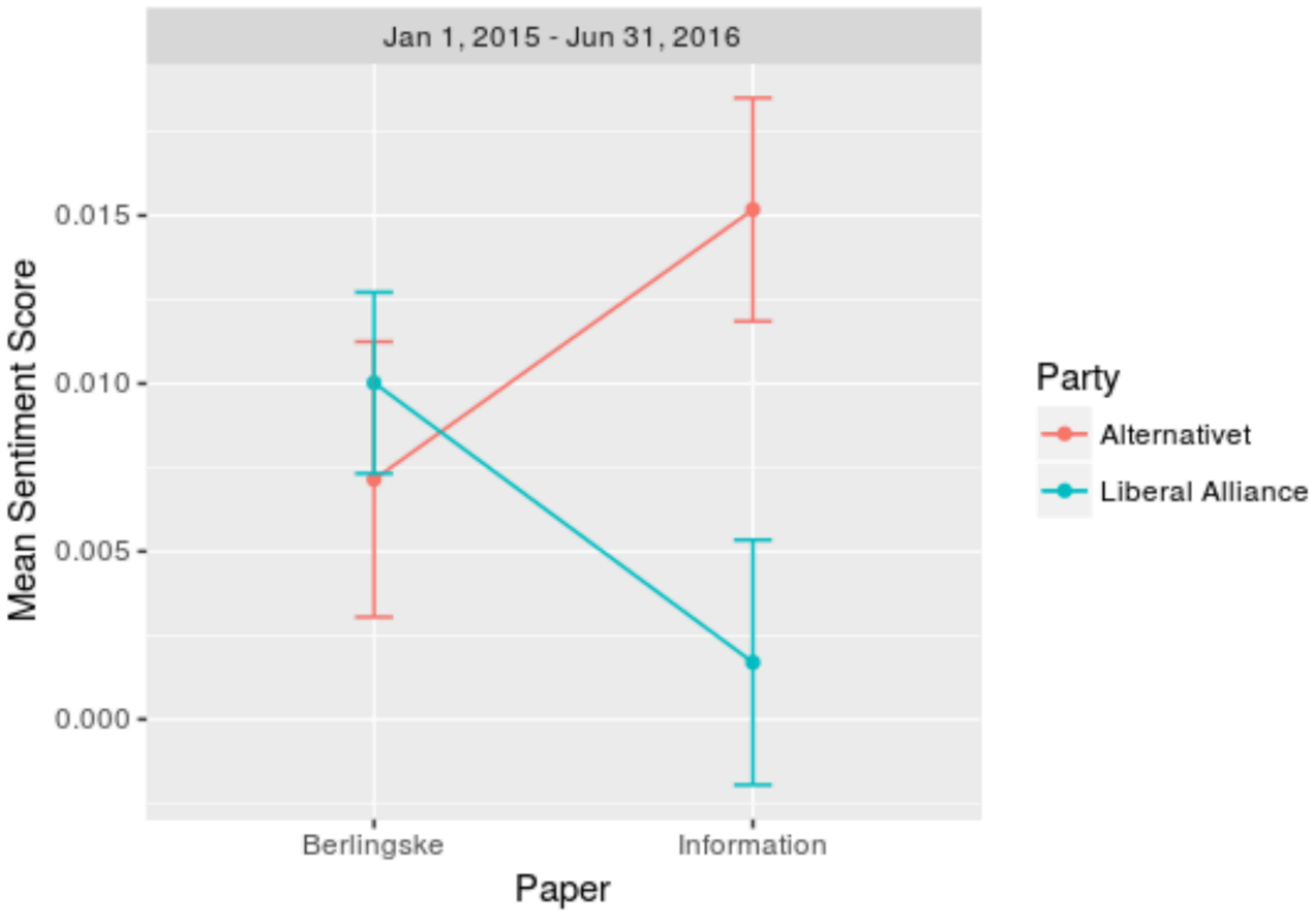


Figure 1: Mean Sentiment Score (MSS) by newspaper and party.

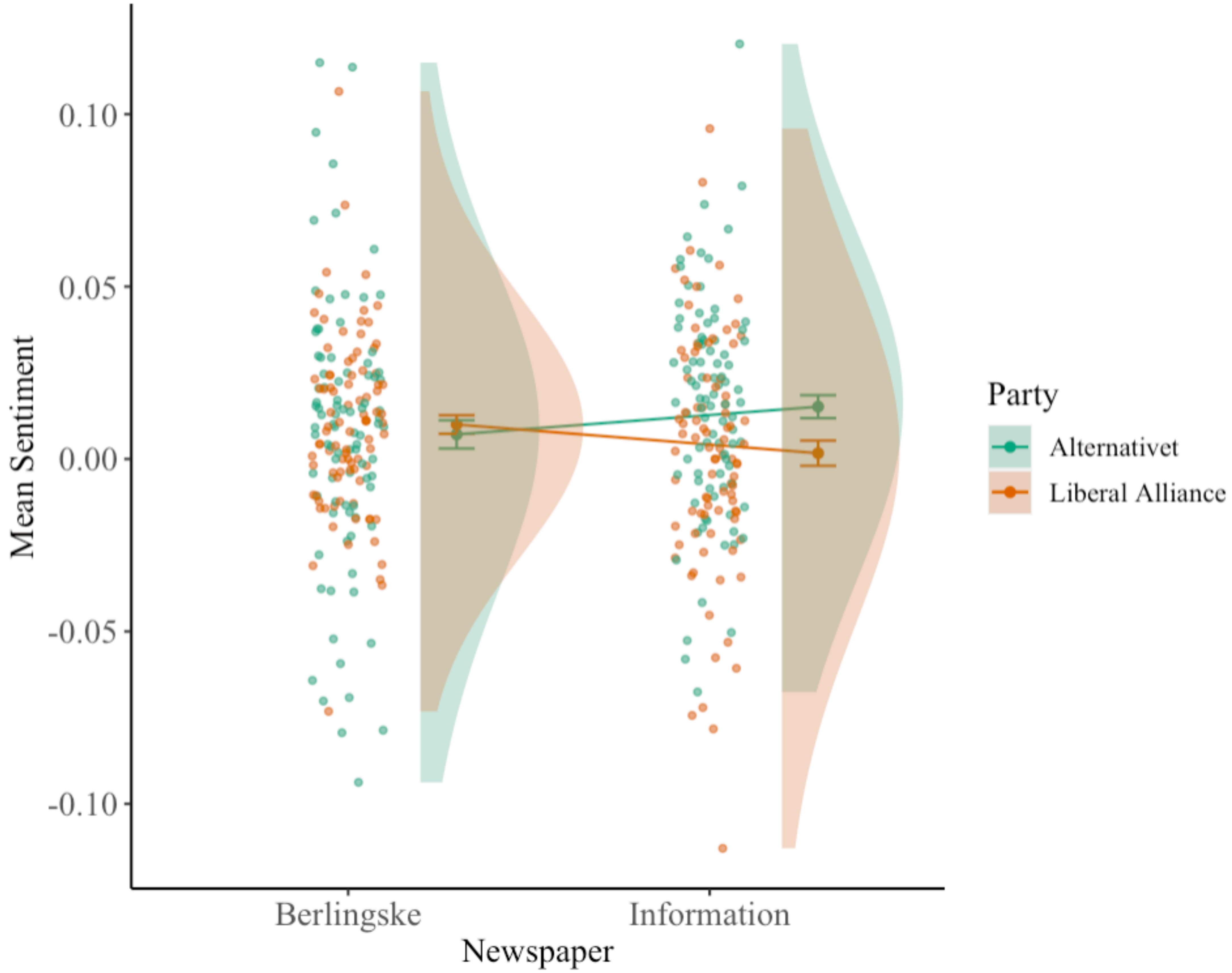


Figure 1.1: Is Danish newspaper really that biases?

Cognitive Load

1. Maximizing Data-ink Ratio
2. Utilize Gestalt Principles
3. Remove superfluous information

Sources: Ferdio, The Visual Display of Quantitative Information

Reductio ad absurdum

Interactive Vizualizations

When more isn't less

Connecting People and Data. Make data pleasant to work with. Happy readers are engaged readers.

Making Systems Playful. Run interactive simulations directly in the browser. No setup required.

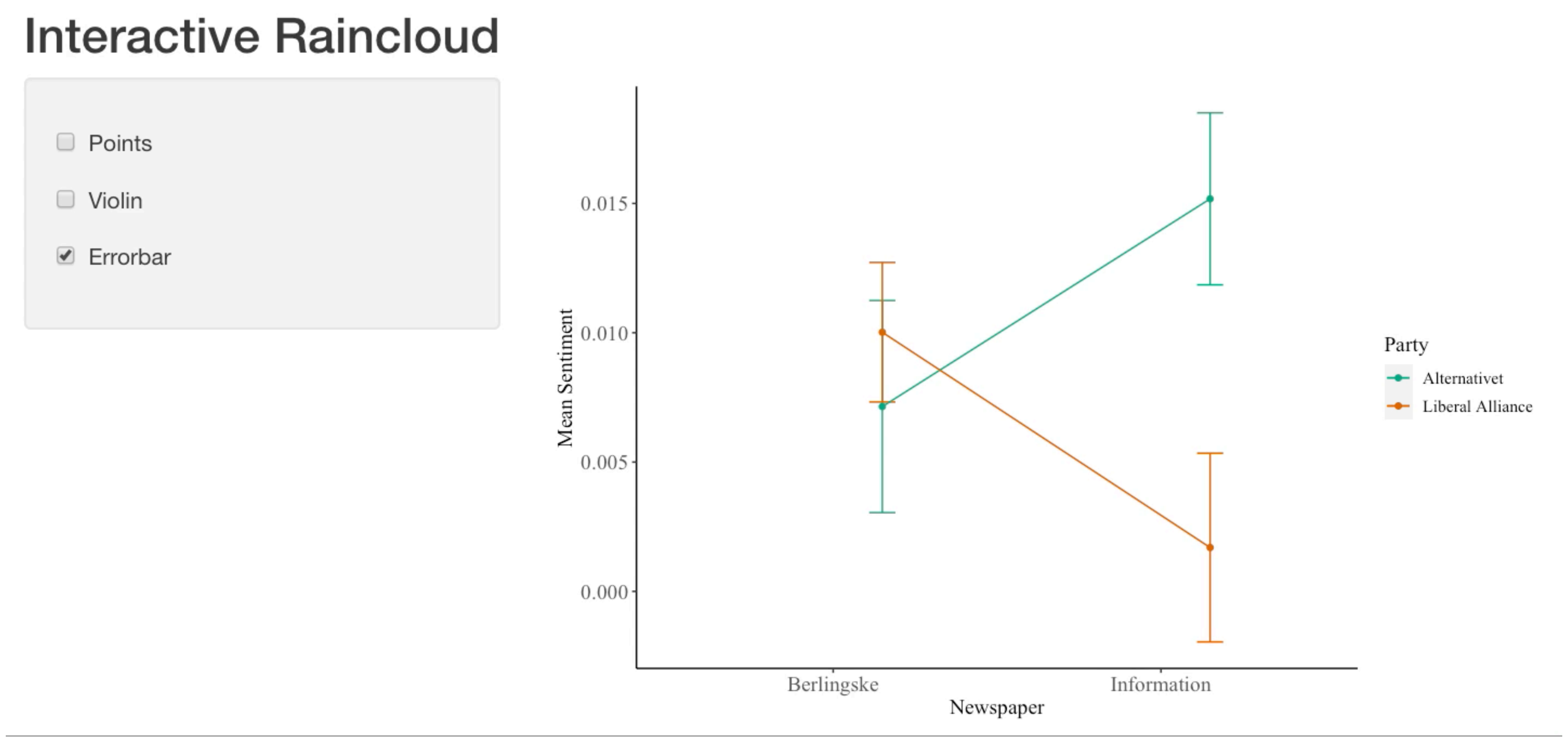
Prompting Self-Reflection. Help readers learn by asking them to reflect in a low pressure environment.

Personalizing Reading. Let readers choose the content that is relevant to their own experience.

Reducing Cognitive Load. Use effective representations to make complex topics more intuitive.

Sources: [Communicating with Interactive Articles, Distill Journal]

An Example



Made using: R Shiny - Could also be made with Streamlit

Interactive Vizualizations

Connecting People and Data.
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Sources: Communicating with Interactive Articles, Distill Journal

The Universal Approximation Theorem in 3 levels of detail.

Readers come with different backgrounds. What if our content could be tailored to their level of knowledge about certain topics?

ILLUSTRATIVE PRECISE

Neural networks can approximate any function that exists. However, we do not have a guaranteed way to obtain such a neural network for every function.

The Universal Approximation Theorem in 3 levels of detail.

Readers come with different backgrounds. What if our content could be tailored to their level of knowledge about certain topics?

ILLUSTRATIVE PRECISE

For any function, there is guaranteed to be a neural network with a finite number (but perhaps a large number) of neurons so that for every possible input, x, the network outputs a close approximation of the function's value f(x). However, we do not have any guarantees on how to train a neural network to learn the correct mapping parameters.

The Universal Approximation Theorem in 3 levels of detail.

Readers come with different backgrounds. What if our content could be tailored to their level of knowledge about certain topics?

ILLUSTRATIVE PRECISE

From mathematical theory of artificial neural networks, the universal approximation theorem states that a feed-forward network with a single hidden layer containing a finite number (but perhaps a large number) of neurons can approximate continuous functions on compact subsets of R^n, as long as the activation function is bounded and continuous. While this says that a simple neural network can represent a wide variety of interesting functions under appropriate parameters, it does not describe how to algorithmically learn such parameters.

Interactive Vizualizations

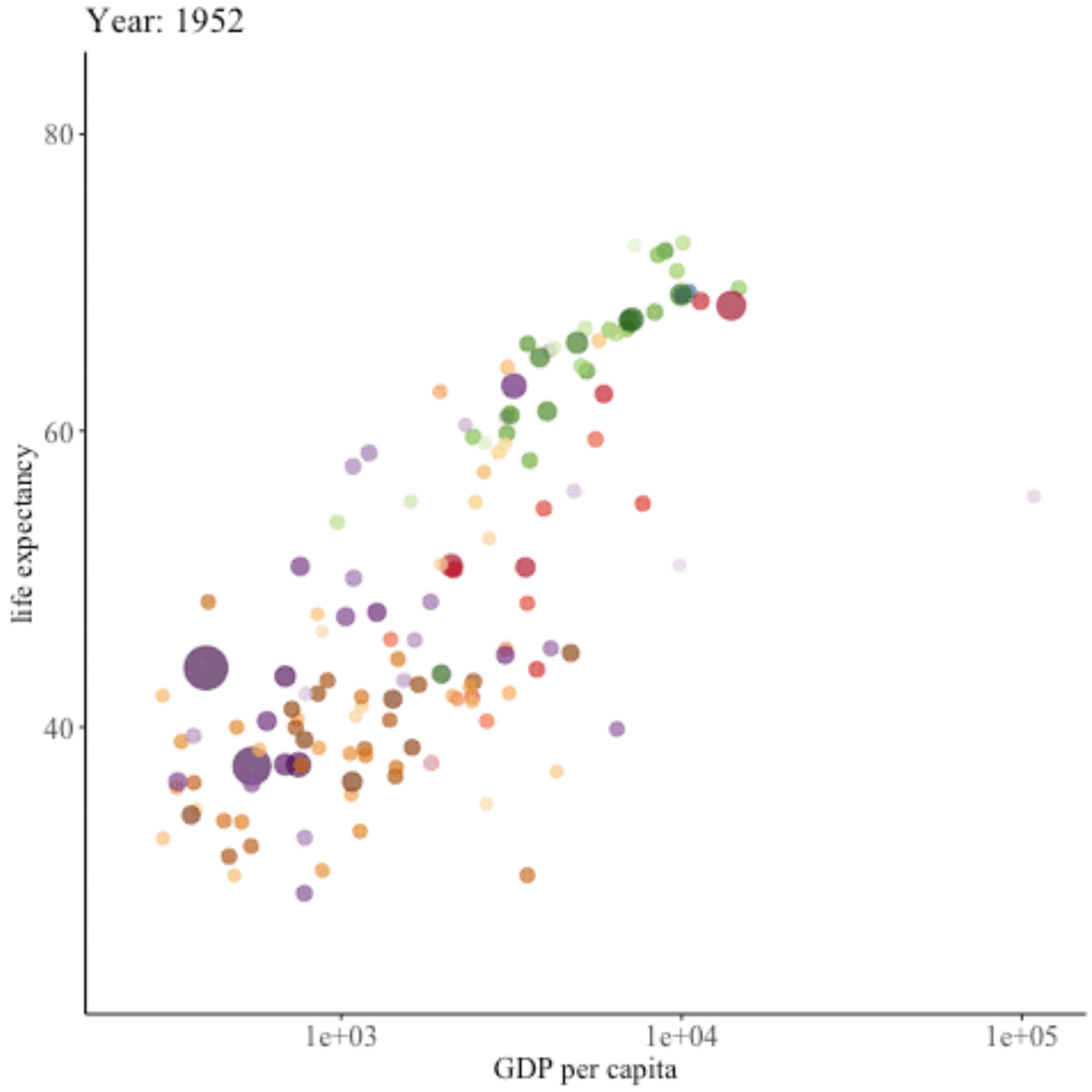
Connecting People and Data. Make data pleasant to work with. Happy readers are engaged readers.	**Making Systems Playful.** Run interactive simulations directly in the browser. No setup required.	**Prompting Self-Reflection.** Help readers learn by asking them to reflect in a low pressure environment.	**Personalizing Reading.** Let readers choose the content that is relevant to their own experience.	**Reducing Cognitive Load.** Use effective representations to make complex topics more intuitive.

Left as an exercise to the reader, mostly Savhannah

Sources: Communicating with Interactive Articles, Distill Journal

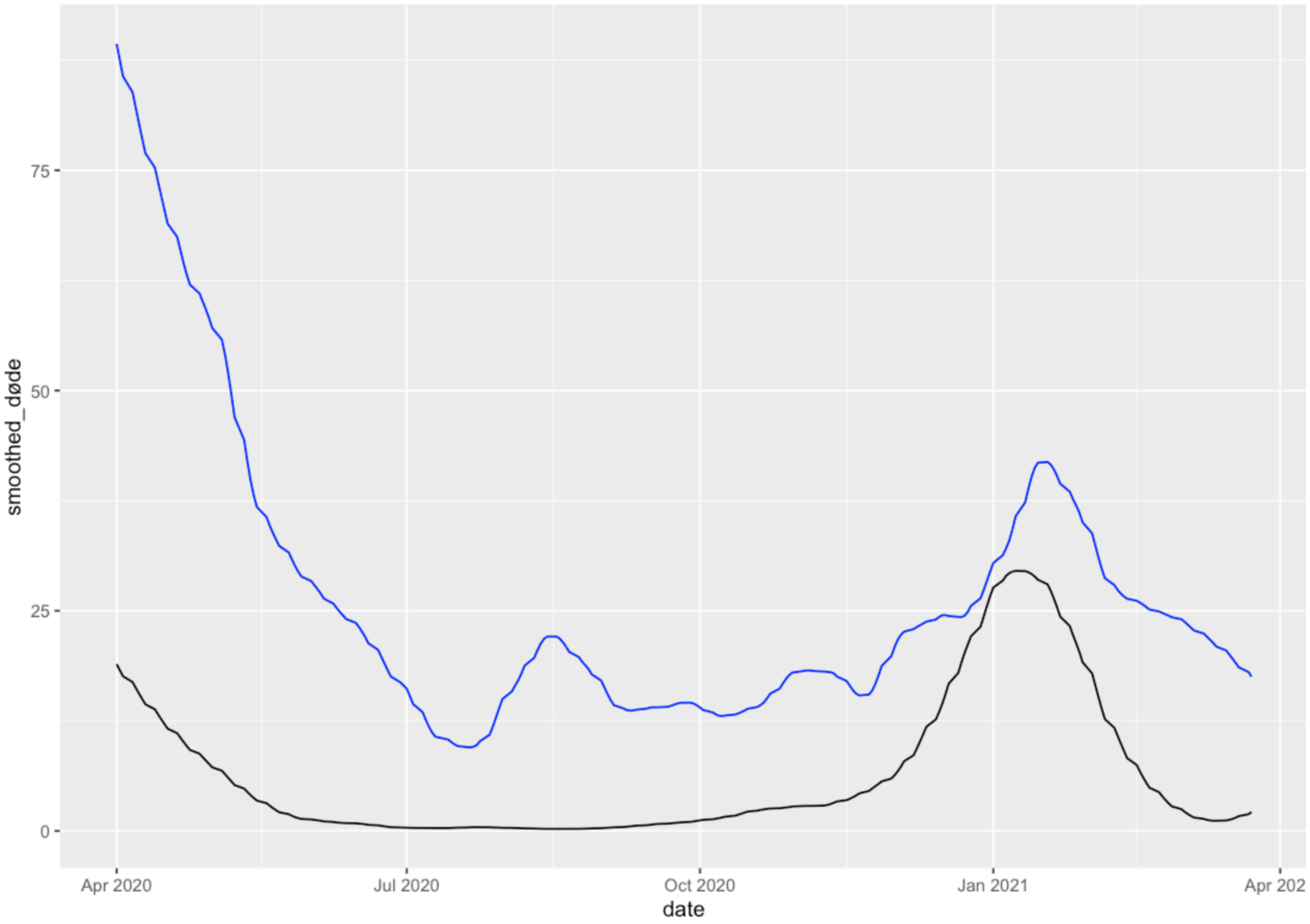
Interactive Vizualizations

A tool for storytelling

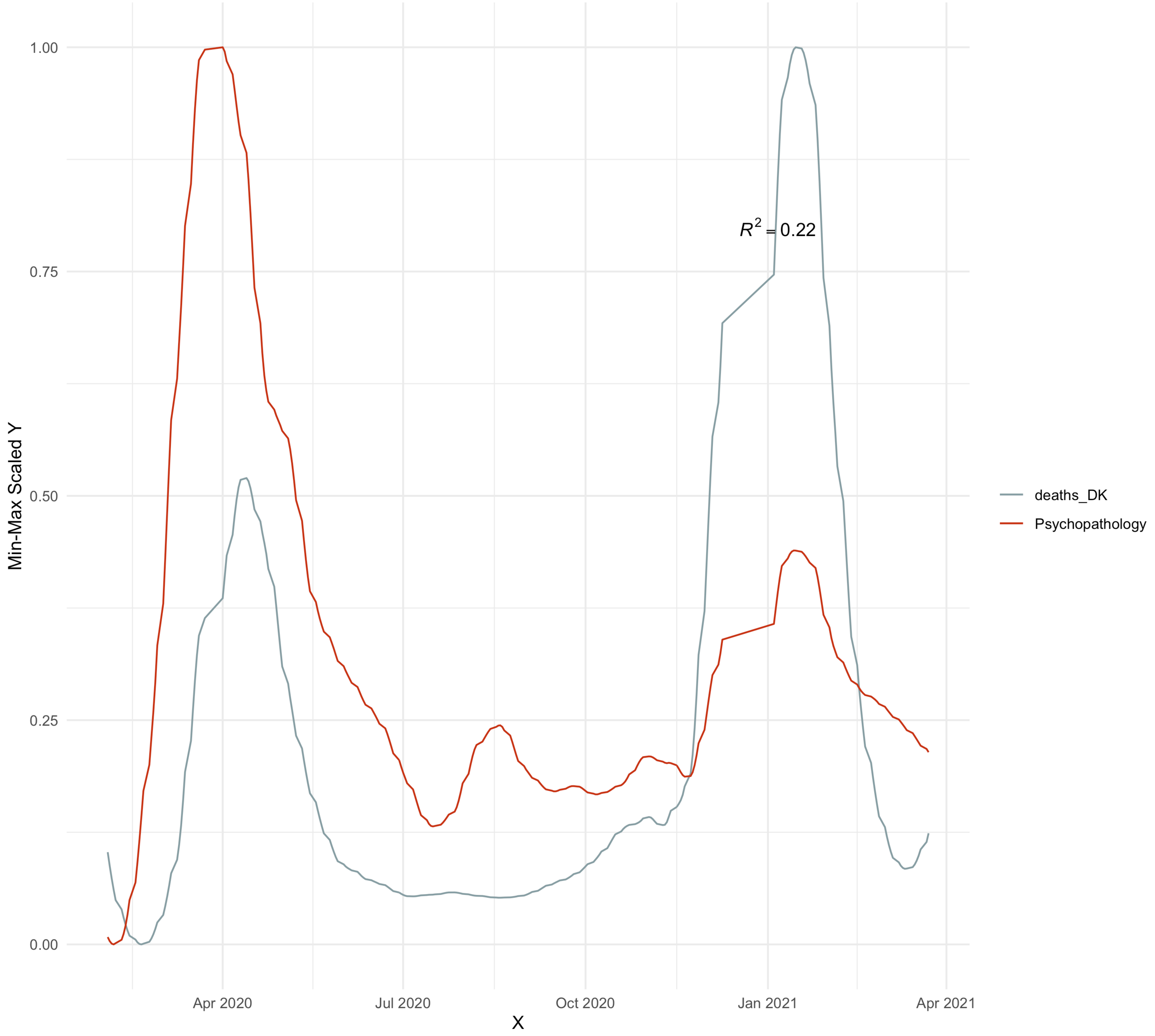


Made using: gganimate - Could also have been made with matplotlib

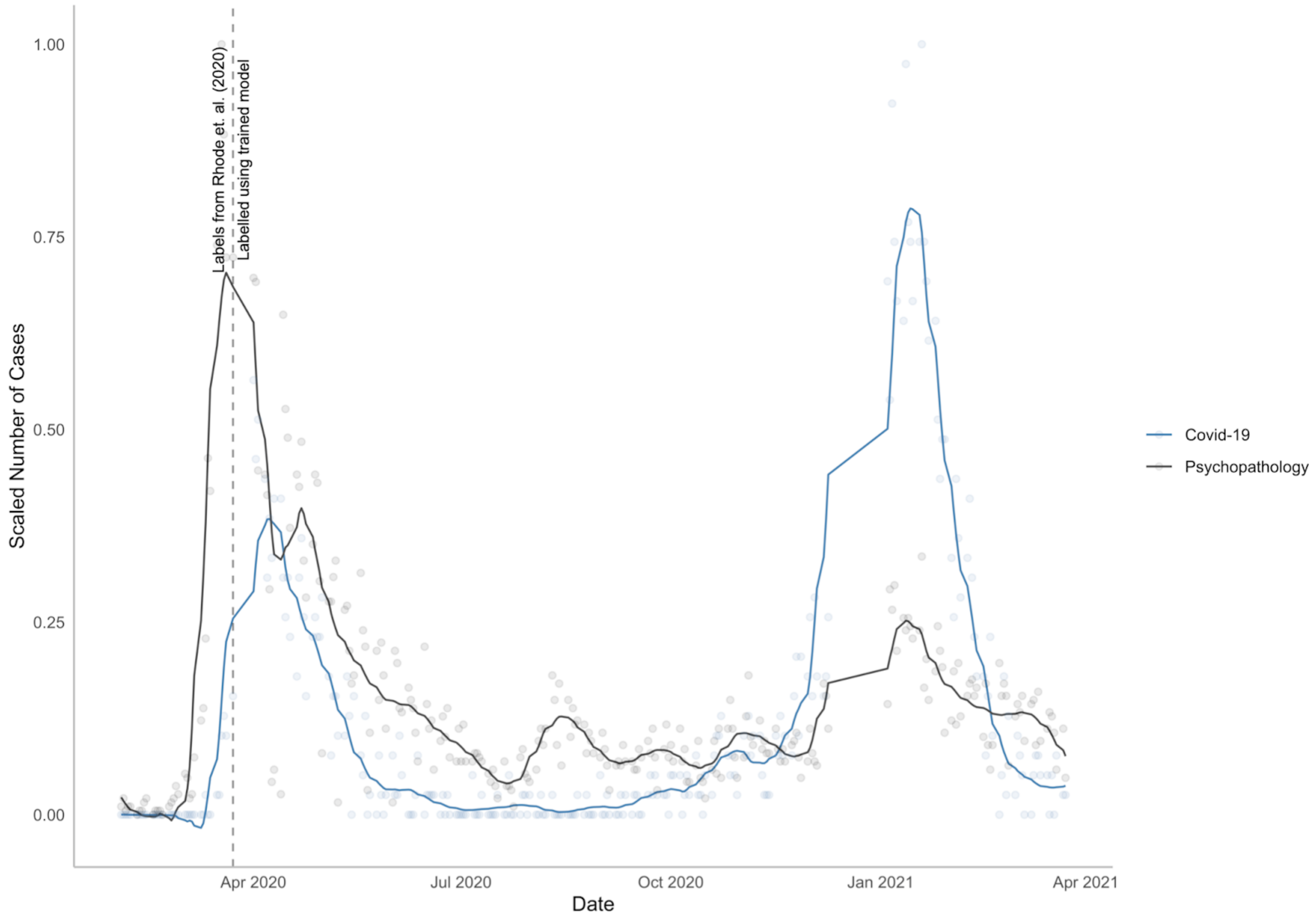
An Example



75
50
25
0
smoothed_døde
Apr 2020
Jul 2020
Oct 2020
Jan 2021
Apr 202
date



$R^2 = 0.22$
Min-Max Scaled Y
1.00
0.75
0.50
0.25
0.00
Apr 2020
Jul 2020
Oct 2020
Jan 2021
Apr 2021
X
deaths_DK
Psychopathology


Labels from Rhode et. al. (2020)
Labelled using trained model
Scaled Number of Cases
1.00
0.75
0.50
0.25
0.00
Apr 2020
Jul 2020
Oct 2020
Jan 2021
Apr 2021
Date
Covid-19
Psychopathology

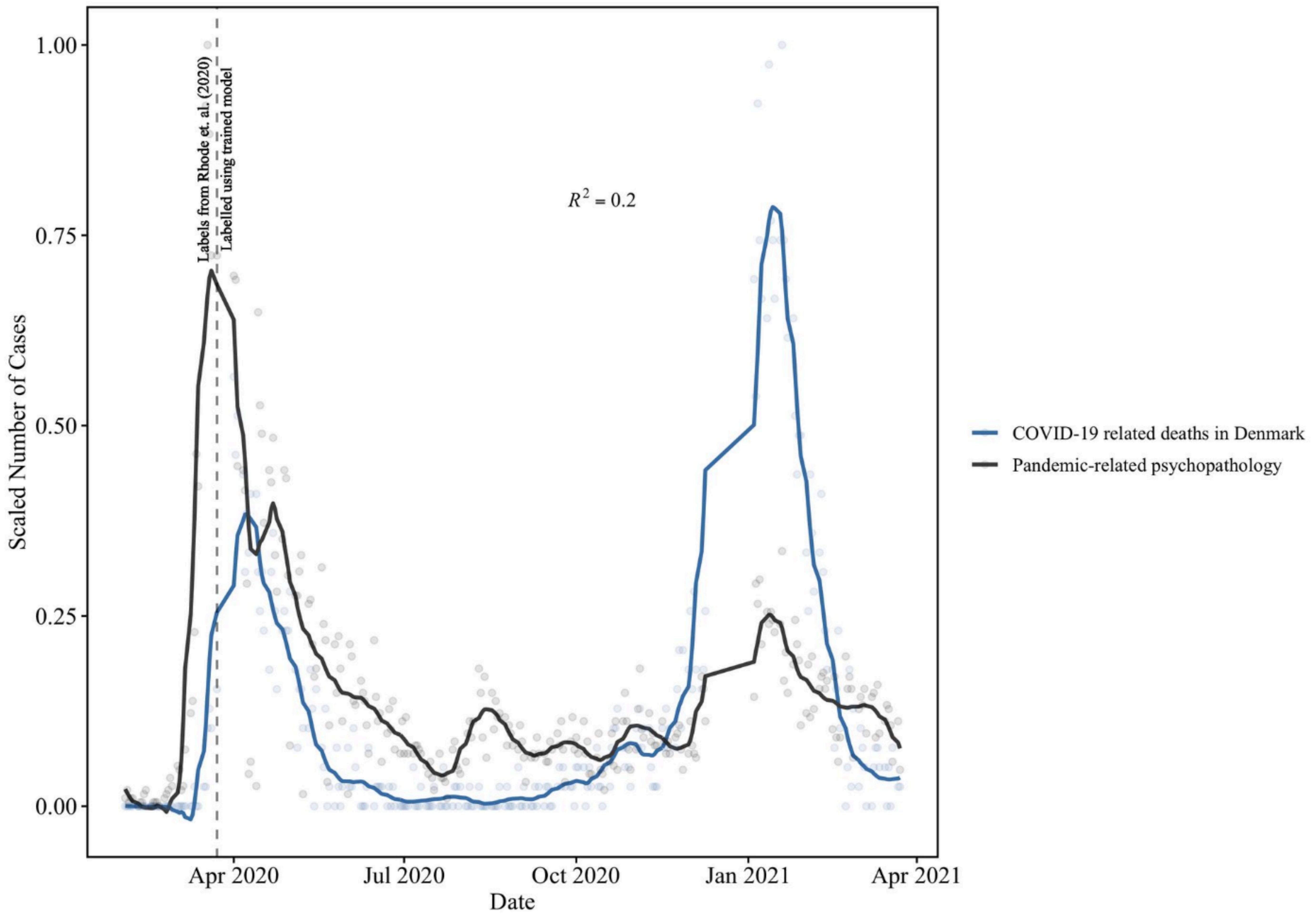
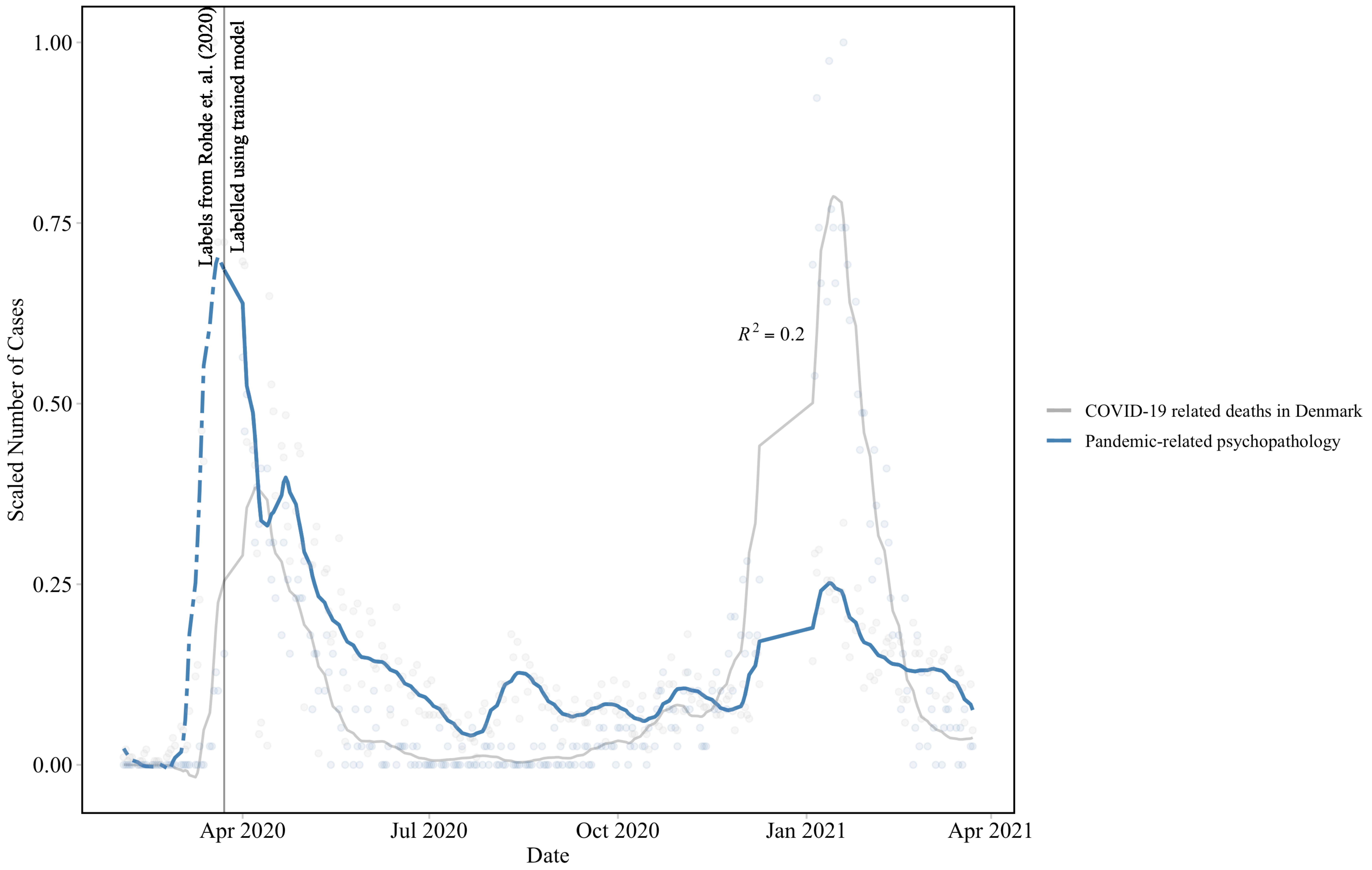


Figure 1: COVID-19 pandemic-related psychopathology and deaths due to COVID-19 over time. Both signals are smoothed using a local polynomial regression with an α=0.12 and holidays and weekends are excluded. The scale is normalized between 0 and 1 to allow for comparison. The dotted line indicates the transition from the clinical notes labeled by Rohde et al. (2020) to the trained model. Data points representing less than 5 cases of pandemic-related psychopathologies are not visualized.



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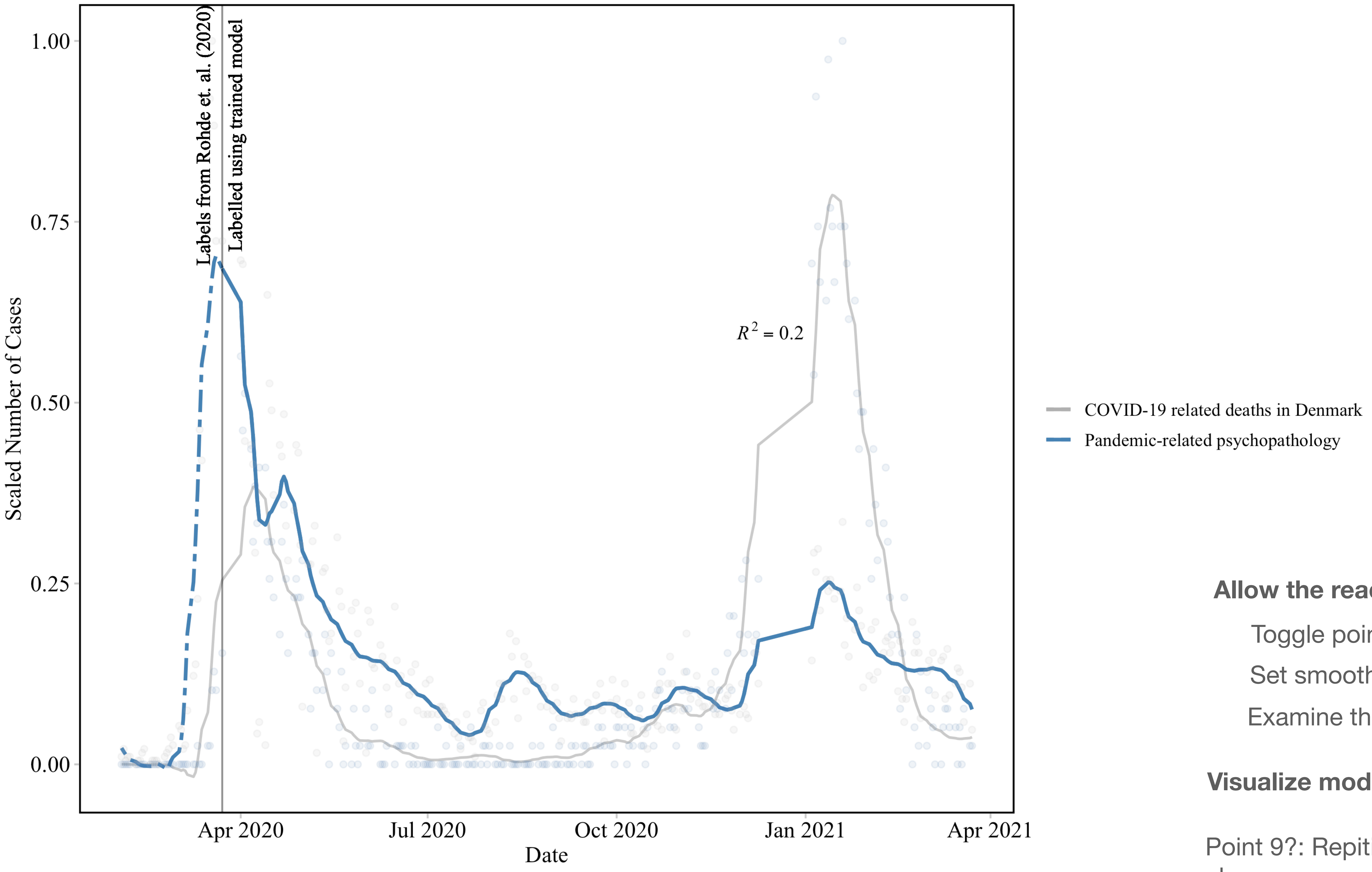

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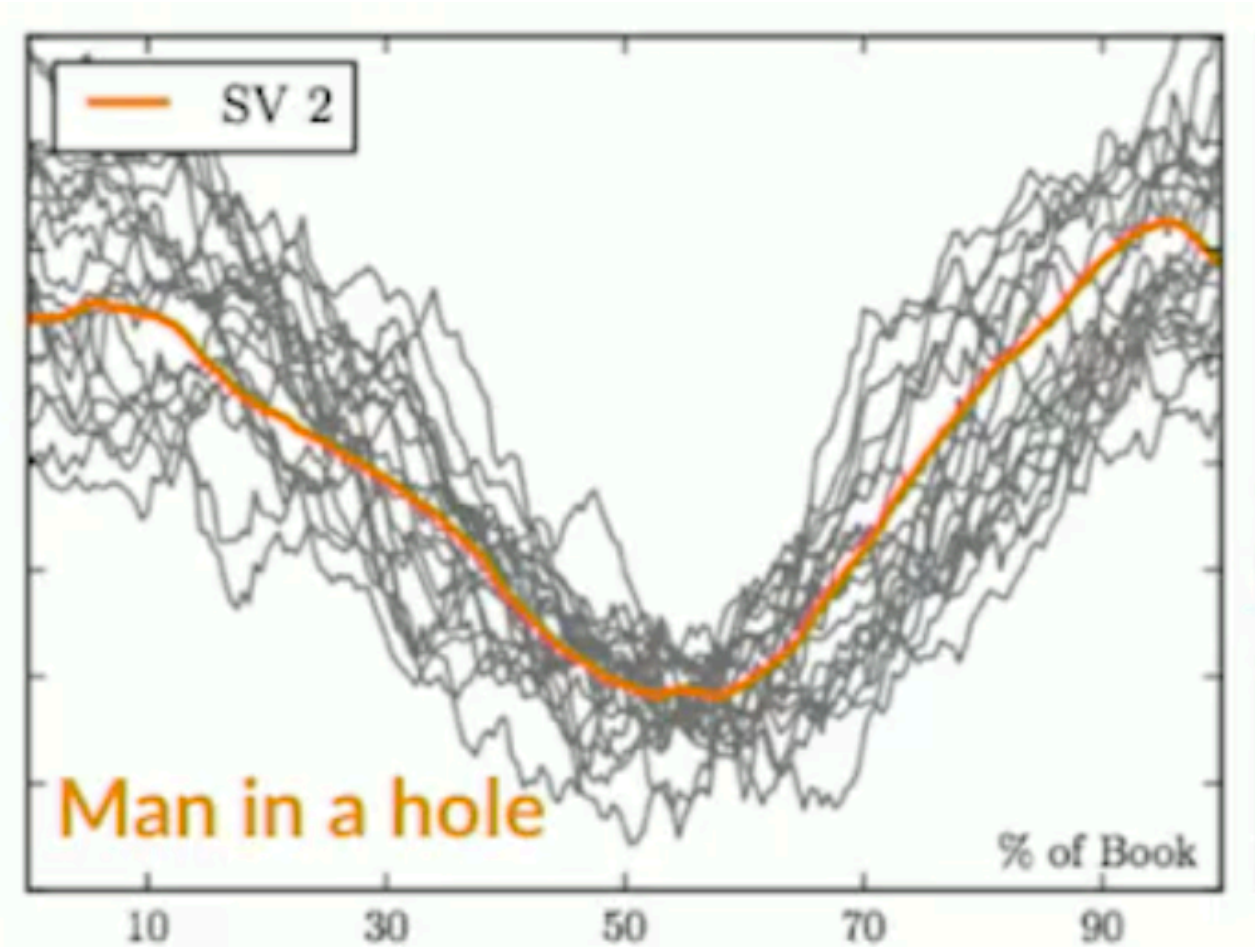
Allow the reader to question the data

- Toggle points
- Set smoothing
- Examine the effect of scaling

Visualize model prediction on text

Point 9?: Repition makes the story clearer

Summary
Interactivity as a compromise between clarity and transparency



Resources

- Online:
  - https://www.ferdio.com/notebook
  - https://thenode.biologists.com/barbarplots/photo/
- Recommended Software
  - Streamlit
  - R Shiny
  - Plotly
  - Bookdown
- Visual storytelling in powerpoints
  - https://www.cliffatkinson.com/beyond-bullet-points
- Article
  - https://distill.pub/2020/communicating-with-interactive-articles/
  - https://link.springer.com/article/10.1140/epjds/s13688-016-0093-1
- Recommended reading:
  - Grammar of graphics
  - Storytelling with Data
  - The design of everyday things
- Visual storytelling in powerpoints
  - https://www.cliffatkinson.com/beyond-bullet-points